

JEE ADVANCED WEEKEND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA



Sec: JR.IIT_*CO-SC(MODEL-A)
Time: 3 Hrs

Date: 30-06-24
Max. Marks: 183

30-06-24_JR.IIT_STAR CO-SC(MODEL-A)_JEE ADV_WAT-12_SYLLABUS

- MATHEMATICS:** **PRESENT WEEK (80%):** Graphs of Trigonometric Functions, Periodicity and extreme values
PREVIOUS WEEK (20%): Conditional Identities, Summation of Trigonometric series, miscellaneous problems on Transformation
- PHYSICS:** **PRESENT WEEK (80%): Circular motion:** Kinematics of Circular Motion, Angular Velocity, Angular Acceleration, Dynamics of Circular Motion, Banking Of Road, Radius of Curvature, Centrifugal Force
(Exclude: Vertical Circular Motion)
PREVIOUS WEEK (20%): FRICTION: Contact Force and its components, Dry Friction, Static and Dynamic Friction, Laws of Friction, Angle of Friction & Repose, Rolling Friction
- CHEMISTRY:** **PRESENT WEEK (80%): Chemical kinetics:** Rates of chemical reactions; Order, Molecularity rate law and Rate constant, Differential and integrated rate expressions for zero and first order reactions, half-life; examples of 1st order Kinetics Types of problems based on first order kinetics, Collision Theory, Arrhenius equation effect of catalyst on rate of reaction, Second order and nth order kinetics, parallel first order reaction
(Exclude: Steady state approximation, sequential, opposing reactions)
PREVIOUS WEEK (20%): Buffer solution & Buffer capacity, Solubility of sparingly soluble salts and solubility product, Indicator, Acid base titration curves

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 7)	Questions With Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – II (Q.N : 8 – 12)	Questions With Integer Answer Type	+3	0	5	15
Sec – III (Q.N : 13 – 18)	Three column paragraph Questions With Single Answer Type	+3	-1	6	18
Total				18	61

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 25)	Questions With Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – II (Q.N : 26 – 30)	Questions With Integer Answer Type	+3	0	5	15
Sec – III (Q.N : 31 – 36)	Three column paragraph Questions With Single Answer Type	+3	-1	6	18
Total				18	61

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 43)	Questions With Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – II (Q.N : 44 – 48)	Questions With Integer Answer Type	+3	0	5	15
Sec – III (Q.N : 49 – 54)	Three column paragraph Questions With Single Answer Type	+3	-1	6	28
Total				18	61

THE NARAYANA GROUP

SECTION – I

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

- Let $f(\theta, n) = \sum_{r=0}^{n-1} \cos^2\left(\theta + \frac{r\pi}{n}\right)$, where $n \in N, n \geq 2$, then which of the following is/are correct
 - $f\left(\frac{\pi}{6}, 10\right) = 5$
 - $f\left(\frac{5\pi}{6}, 10\right) = \frac{5}{2}$
 - $f\left(\frac{\pi}{2}, 14\right) = 0$
 - $f(\pi, 14) = 7$
- If $u = \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} + \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$ then which of the following is/are TRUE? ($\forall \theta \in R$)
 - $u_{\max}^2 = 2(a^2 + b^2)$
 - $u_{\max}^2 = (a^2 + b^2)$
 - $u_{\min}^2 = (a + b)^2$
 - $u_{\max}^2 - u_{\min}^2 = (a - b)^2$
- Which of the following is periodic ([.] is G.I.F) having fundamental period
 - $f(x) = \sin\left(\frac{\pi}{2}[x]\right)$
 - $h(x) = \frac{\sin^2 x + \cos^4 x}{\cos^2 x + \sin^4 x}$
 - $g(x) = \sin(\sqrt{2}x) + \cos(\sqrt{2}x)$
 - $\phi(x) = \sin(\pi[x]) + [x+1] - x$
- Let $f(x) = ab \sin x + b\sqrt{1-a^2} \cos x + c$, where $|a| < 1, b > 0$ then
 - maximum value of $f(x)$ is b if $c = 0$
 - difference of maximum and minimum values $f(x)$ is $2b$
 - fundamental period of $f(x)$ is 2π
 - fundamental period of $f(x)$ is π

5. Which of the following curves does not intersect $y = 1$?

A) $y = |\operatorname{cosec} x + 1|$

B) $y = \sqrt{|\tan x|}$

C) $y = |\sin x + 4|$

D) $y = 2 \sin 3x$

6. Let

$$f_n(\theta) = 2 \sin \frac{\theta}{2} \sin \frac{3\theta}{2} + 2 \sin \frac{\theta}{2} \sin \frac{5\theta}{2} + 2 \sin \frac{\theta}{2} \sin \frac{7\theta}{2} + \dots + 2 \sin \frac{\theta}{2} \sin (2n+1) \frac{\theta}{2}, n \in N,$$

Then Which value of the following is / are correct?

A) $f_9\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

B) $f_n\left(\frac{2\pi}{n}\right) = 0, n \in N$

C) $f_5\left(\frac{2\pi}{7}\right) = 0$

D) $f_9\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$

7. Let $f(x) = \frac{\sin x}{\sqrt{1-\cos^2 x}} + \frac{\cos x}{\sqrt{1-\sin^2 x}} + \frac{\tan x}{\sqrt{\sec^2 x - 1}} + \frac{\cot x}{\sqrt{\operatorname{cosec}^2 x - 1}}$, wherever $f(x)$ is defined, which of the following is correct.

A) maximum value of $f(x)$ is 4.

B) minimum value of $f(x)$ is -2.

C) Range of $f(x) = \{4\}$

D) Range of $f(x) = \{-2, 0, 4\}$

SECTION-II

(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

8. Let $P = (4 \tan 1^\circ)(4 \tan 2^\circ)(4 \tan 3^\circ) \dots (4 \tan 10^\circ)$,

$$Q = (\tan 2^\circ)(\tan 4^\circ)(\tan 6^\circ) \dots (\tan 20^\circ) \text{ and}$$

$$R = (1 + \tan 1^\circ)(1 - \tan 1^\circ)(1 + \tan 2^\circ)(1 - \tan 2^\circ) \dots (1 + \tan 10^\circ)(1 - \tan 10^\circ). \text{ If } P = \lambda QR,$$

then the value of $\frac{\lambda}{128}$ is equal to

9. The number of solutions of the equation $\sin\left(\frac{x}{2}\right) + 2\pi x = x^2 + \pi^2 + 1$ is,

10. If the fundamental period of the function $f(x) = \sin\left(\frac{\pi[x]}{12}\right) + \tan\left(\frac{\pi[x]}{3}\right) + \cos\left(\frac{\pi[x]}{4}\right)$ is λ (where $[.]$ is G.I.F, then the value of $\frac{3\lambda}{8}$ is equal to
11. If $\theta_1, \theta_2, \theta_3, \dots, \theta_{10} \in \left[0, \frac{\pi}{2}\right]$ such that $\sin^2 \theta_1 + \sin^2 \theta_2 + \dots + \sin^2 \theta_{10} = 1$ then least value of $\frac{\cos \theta_1 + \cos \theta_2 + \dots + \cos \theta_{10}}{\sin \theta_1 + \sin \theta_2 + \dots + \sin \theta_{10}}$ is equal to
12. If the fundamental period of the function $f(x) = 4\cos^4\left(\frac{x-\pi}{4\pi^2}\right) - 2\cos\left(\frac{x-\pi}{2\pi^2}\right)$ is equal to $k\pi^3$, then the value of 'k' is equal to

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q,13, Q,14 and Q,15 by appropriately matching the information given in the three columns of the following table.

Column 1,2,3 contains trigonometric function, maximum value of the function, minimum value of the function.

COLUMN-1	COLUMN-2	COLUMN-3
(I) $f(x) = \frac{7 + 6 \tan x - \tan^2 x}{1 + \tan^2 x}$ For all values of $x \neq (2n+1)\frac{\pi}{2}$	(i) maximum value of $f(x)$ is equal to 10	(P) minimum value of $f(x)$ is 2
(II) $f(x) = 5 \cos x + 3 \cos\left(x + \frac{\pi}{3}\right) + 3$ For all real values of x	(ii) maximum value of $f(x)$ is 8	(Q) minimum value of $f(x)$ is 7
(III) $f(x) = \sqrt{9 \sin^2 x - 30 \sin x + 25}$ For all real values of x	(iii) maximum value of $f(x)$ is 6	(R) minimum value of $f(x)$ is -2
(IV) $f(x) = \sqrt{(3 \sin x - 4 \cos x - 10)(3 \sin x + 4 \cos x - 10)}$ For all real values of x	(iv) maximum value of $f(x)$ is 13	(S)) minimum of value $f(x)$ is -4

13. Which of the following options is the only **CORRECT** combination
- A) (I) (ii) (S) B) (II) (i) (P) C) (IV) (iv) (Q) D) (III) (i) (P)
14. Which of the following options is the only **CORRECT** combination
- A) (II) (i) (S) B) (I) (i) (P) C) (III) (ii) (S) D) (IV) (ii) (Q)
15. Which of the following options is the only **INCORRECT** combination
- A) (II) (i) (S) B) (III) (ii) (P) C) (IV) (iv) (Q) D) (I) (ii) (Q)

Answer Q,16, Q,17 and Q,18 by appropriately matching the information given in the three columns of the following table.

Here $[.]$ represents greatest integer function, $\{.\}$ denote fractional part function:

Column-1	Column-2	Column-3
(I) Fundamental period of $e^{\cos^4 \pi x} + x - [x] + \cos^2 \pi x$ is	(i) Rational number	(P) π
(II) Fundamental period of $\cos(2\pi\{2x\}) + \sin(2\pi\{2x\})$ is	(ii) Irrational number	(Q) 2π
(III) Fundamental period of $\cos(\sin(x + \sin x))$ is	(iii) Natural number	(R) $\frac{1}{2}$
(IV) Fundamental period of $\left \cos^5\left(\frac{x}{2}\right) \right + \sqrt{\sin^{10}\left(\frac{x}{2}\right)}$ is	(iv) Prime number	(S) 1

16. Which of the following options is the only **INCORRECT** combination
- A) (I) (i) (S) B) (II) (i) (R) C) (I) (iii) (S) D) (IV) (ii) (Q)
17. Which of the following options is the only **CORRECT** combination
- A) (II) (i) (R) B) (III) (i) (S) C) (IV) (ii) (Q) D) (I) (ii) (S)
18. Which of the following options is the only **CORRECT** combination
- A) (I) (i) (R) B) (II) (ii) (P) C) (III) (ii) (Q) D) (IV) (iii) (S)

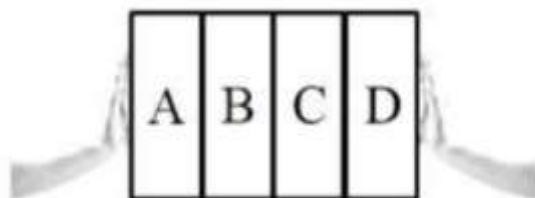
SECTION – I

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

19. Four identical blocks are held in vertical position as shown.



Each block has mass m , coefficient of friction between hands and books A and D is $\frac{\mu}{2}$, between C and D, B and C, A and B is $\frac{\mu}{6}$ each. If each hand exerts a force F on the books in contact, Then

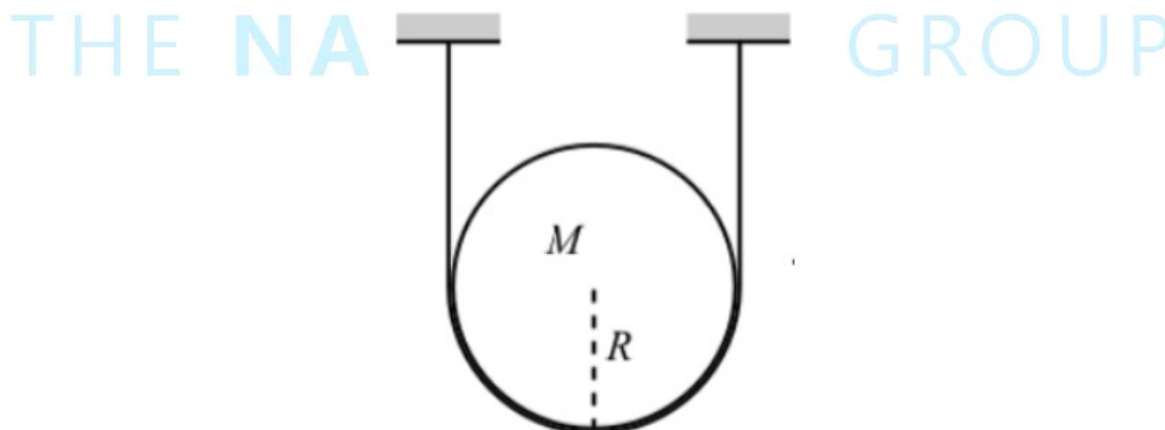
A) The minimum force applied in order to prevent slipping anywhere is $\frac{6mg}{\mu}$

B) If $F = \frac{4mg}{\mu}$, there will be no slipping between hand and A

C) For any value of F there is no slipping between B and C

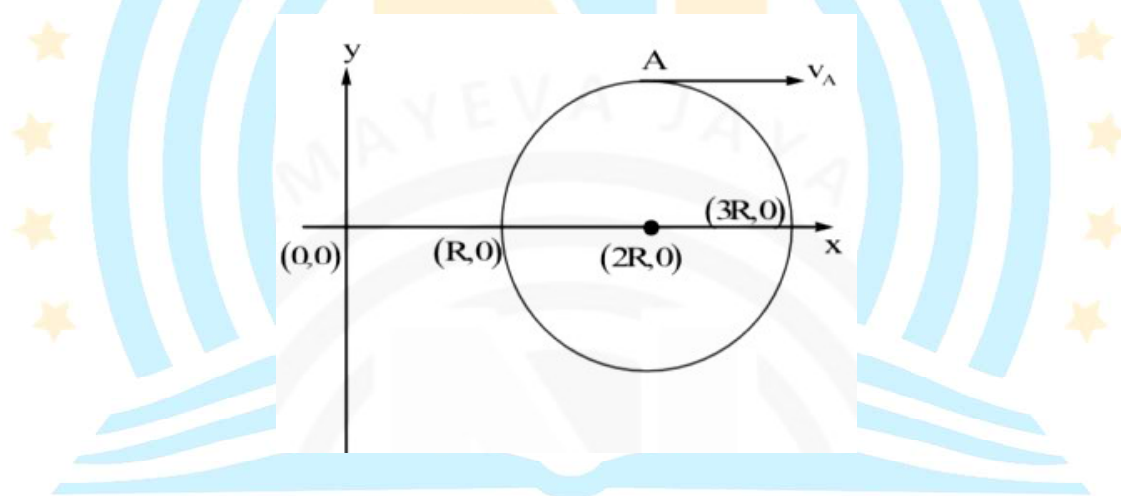
D) If $F = \frac{10mg}{\mu}$, there will be no slipping between C and D

20. A disc of mass m and radius R is held up by a string as shown in figure. Then



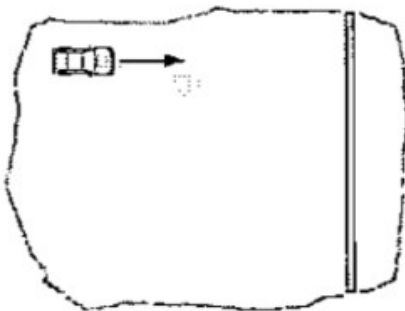
- A) If string is massless & disc is frictionless then tension in string is $\frac{mg}{2}$
- B) If string is massless and coefficient of friction between disc & string is μ then smallest possible tension in the string at its lowest point is $\frac{mg}{2}e^{-0.5\mu\pi}$
- C) If string is massless and coefficient of friction between disc & string is μ then largest possible tension in the string at its lowest point is $\frac{mg}{2}$
- D) If string is massive with constant mass density $\lambda(\text{kg/m})$ and disc is frictionless then tension at lowest point is $\frac{mg}{2} + \frac{\lambda\pi Rg}{2} - \lambda Rg$

21. A particle is executing uniform circular motion of angular velocity ω (about center of circle) in xy plane with radius R and center $(2R, 0)$. When it is at point A, answer the following questions

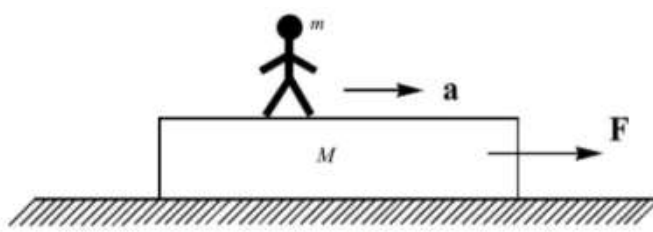


- A) Transverse acceleration of particle wrt origin is $\frac{2R\omega^2}{\sqrt{5}}$
- B) Radial acceleration of particle wrt origin is $R\omega^2$
- C) Magnitude of angular acceleration of particle wrt origin is $\frac{6\omega^2}{25}$
- D) Angular velocity of particle about origin is $\frac{\omega}{5}$

22. A car is moving with speed 108 km/h on a large uniform horizontal pavement perpendicular towards a wall. To avoid collision, the driver takes a turn and finally runs the car parallel to the wall with the same speed. Coefficient of friction between the tyres and the pavement is 0.6, acceleration of free fall is 10 m/s^2 and all the load of the car is concentrated on the front axle (means you can assume that limiting friction acting on car is μmg , where m is total mass of car). If the turning process is the fastest one on this pavement, check validity of the following statements.



- A) The car follows a circular path during the turn
 B) The car follows a parabolic path during the turn
 C) Minimum radius of curvature of the path is 75 m.
 D) Initial distance of the wall must be greater than $75\sqrt{2} \text{ m}$
23. A car of mass M is moving on a horizontal circular path of radius r . At an instant its speed is v and is increasing at a rate a .
- A) The acceleration of the car is towards the centre of the path.
 B) The magnitude of the frictional force on the car is greater than $\frac{mv^2}{r}$.
 C) The friction coefficient between the ground and the car is not less than a/g .
 D) The friction coefficient between the ground and the car is $\mu = \tan^{-1} \frac{v^2}{rg}$
24. A plank of mass M is placed on a rough horizontal surface. A man of mass m walks on the plank with an acceleration ' a ' while the plank is also acted upon by a horizontal force F whose magnitude and direction can be adjusted to keep the plank at rest. Coefficient of friction between plank and surface is μ . Choose the correct option(s)



- A) The maximum value of F to keep the plank at rest is $ma + \mu(M + m)g$
- B) The minimum value of F to keep the plank at rest is $ma - \mu(M + m)g$
- C) Direction of friction on the plank due to ground will always be towards right
- D) If $F = ma$ then friction on man due to plank is also F and friction between plank and ground is zero
25. A person applies a constant force $\vec{F}$ on a particle of mass m and finds that the particle move in a circle of radius r with a uniform speed v .
- A) This is not possible
- B) There are other forces also on the particle
- C) The resultant of other forces is mv^2/r towards centre
- D) The resultant of the other forces varies in magnitude as well as direction

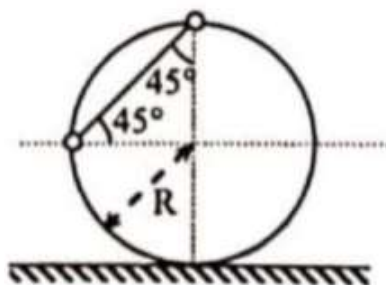
SECTION-II
(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

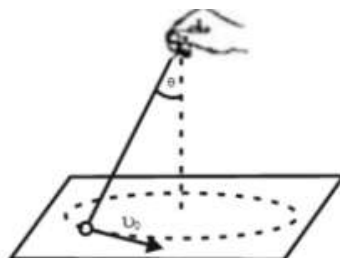
Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases.

26. A particle of mass 1 kg is acted upon by a constant tangential force of 1N and a normal force of magnitude $2v^2$ where v is the speed of the particle at any instant. If the particle starts from rest, then find the angle (in radian) turned by velocity vector in 1s.
27. Two beads connected by a light inextensible string are placed over fixed vertical ring as shown in figure. If mass of each bead is 100 gms and all surface are frictionless if T is the tension in the string just after the beads are released from the shown position, then find the value of $8T^2$ (in N^2) ?

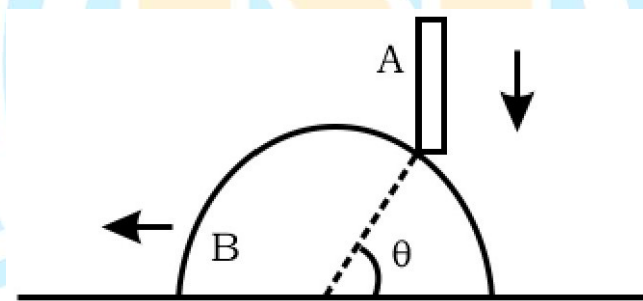
If it is $\frac{1}{x}$. Find the value of x . (take $g = 10m/s^2$)



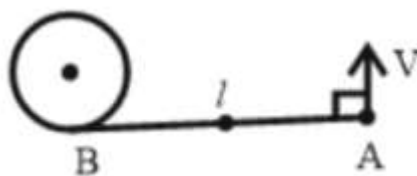
28. One end of light inextensible thread of length $\ell = 1\text{m}$ is held stationary over a frictionless horizontal floor, while a small bead tied to the other end of the thread is describing a circular path with a small speed $v_0 = 1\text{m/s}$ on the floor as shown in figure. The upper end of the thread is suddenly pulled vertically upwards with a constant velocity $v = 1\text{ m/s}$. If bead does not leave the floor, then immediately after the upper end is pulled (take $\theta = 30^\circ$), if the radius of curvature (in meter) of the path traced by the bead is R , then find the value of $10 R$.



29. A rod A which is constrained to move along vertical only (the required forces are not shown). Find the value of acceleration of wedge B at the moment when acceleration and velocity of A are $\frac{8}{\sqrt{3}}\text{m/s}^2$ and 1 m/s , both vertically downward. Also given that $R = 1\text{ m}$ and $\theta = 60^\circ$ at that same moment.



30. A horizontal plane supports a stationary vertical cylinder of radius R , and a particle A attached to the cylinder by a horizontal thread AB of length l as shown in the figure (top view). Now the particle is imparted a velocity v perpendicular to the thread



Then acceleration of mid point of thread just after imparting velocity is given as $x \frac{v^2}{\ell}$.

Find the value of $2x$. (Given that $R = 2\sqrt{6}l$ and ignore gravity)

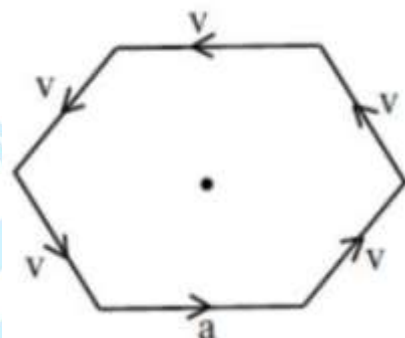
SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q,31, Q,32 and Q,33 by appropriately matching the information given in the three columns of the following table.

Six particles are located at the vertices of regular hexagon of side 'a'. At $t = 0$ they all start moving simultaneously with constant speed 'V'. The first particle is heading for second, the second particle is heading for the third and so on.



Velocity v , initial side a and instantly t are given in the three column's respectively.

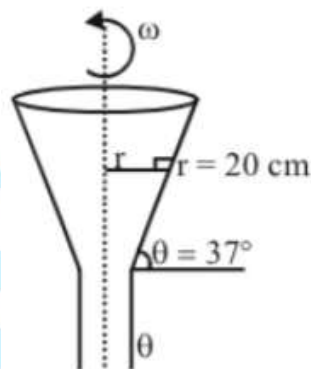
Column – 1		Column – 2		Column – 3	
I)	$v = \sqrt{3} \text{ m/s}$	i)	$a = 4 \text{ m}$	p)	$t = \frac{1}{3} \text{ s}$
II)	$v = 1 \text{ m/s}$	ii)	$a = 1 \text{ m}$	q)	$t = 1 \text{ s}$
III)	$v = 2\sqrt{3} \text{ m/s}$	iii)	$a = 2 \text{ m}$	r)	$t = \frac{1}{2\sqrt{3}} \text{ s}$
IV)	$v = 3 \text{ m/s}$	iv)	$a = 1.5 \text{ m}$	s)	$t = 2 \text{ s}$

31. In which case angular velocity of a particle about centroid of hexagon is 2 rad/s .
A) II, i, s B) I, ii, R C) II, i, Q D) III, ii, P
32. In which case acceleration of particle is $4\sqrt{3} \text{ m/s}^2$.
A) I, iii, Q B) I, ii, P C) III, iii, R D) III, ii, P
33. In which case radius of curvature is $\frac{7}{\sqrt{3}} \text{ m}$.
A) II, i, Q B) I, ii, P C) I, iii, P D) I, ii, Q

Answer Q,34, Q,35 and Q,36 by appropriately matching the information given in the three columns of the following table.

A very small cube of mass 2kg is placed on the surface of a funnel as shown in figure.

The funnel is rotating about its vertical axis of symmetry with angular velocity ' ω '. The wall of funnel makes an angle 37° with horizontal. The distance of cube from the axis of rotation is 20 cm and frictional coefficient is μ . (Take $g = 10\text{m/s}^2$)



The values of ω , μ & frictional forces are given in the 3 columns respectively.

Column – 1		Column – 2		Column – 3	
I)	$\sqrt{10}$ rad/s	i)	0.2	P)	4.16 N
II)	2 rad/s	ii)	0.1	Q)	1.84 N
III)	$\sqrt{20}$ rad/s	iii)	0.3	R)	3N
IV)	1 rad/s	iv)	0.4	S)	4.87 N

34. Which of the following is correct.

- A) II, i, P B) I, ii, Q C) III, i, Q D) IV, ii, R

35. Which of the following is correct.

- A) III, i, p B) I, ii, Q C) I, iii, R D) II, i, R

36. Which of the following is correct

- A) I, i, P B) I, ii, Q C) IV, iii, R D) IV, iii, S

SECTION – I

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

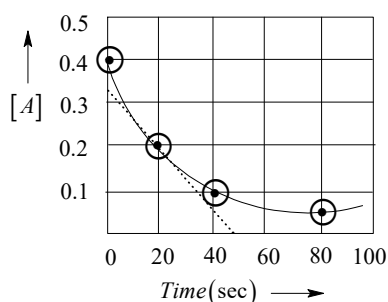
37. The rate of reaction between hemoglobin (*Hb*) and *CO*, $4Hb + 3CO \longrightarrow Hb_4(CO)_3$ has been studied at 20°C. Concentrations are expressed in micromoles per liter (μ mol/L).

Concentration (μ mol/L)		Rate of reaction (μ mol L ⁻¹ s ⁻¹)
<i>Hb</i>	<i>CO</i>	
3.36	1.00	0.941
6.72	1.00	1.88
6.72	3.00	5.64

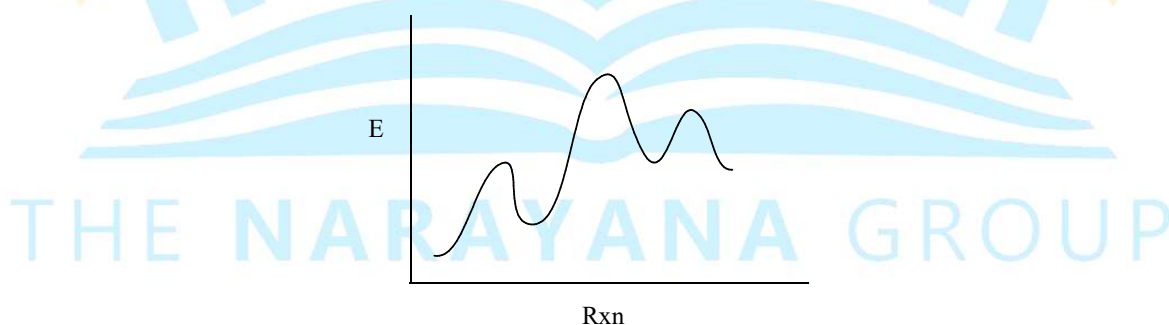
The conclusion(s) that can be made from this observation is/are

- A) The rate constant of the reaction is $5.25 \text{ L mol}^{-1} \text{ s}^{-1}$.
- B) The overall order of the reaction is 2.
- C) When the concentration of *Hb* is $16.8 \mu \text{ mol/L}$ and *CO* is $2.00 \mu \text{ mol/L}$, then the rate is $9.41 \mu \text{ mol L}^{-1} \text{ s}^{-1}$.
- D) This reaction is an elementary reaction.
38. Which of the following statements is/are true?
- A) The rate of formation of NO is $3.6 \times 10^{-3} \text{ M s}^{-1}$ in $4NH_{3(g)} + 5O_{2(g)} \rightarrow 4NO_{(g)} + 6H_2O_{(g)}$ then the rate of formation of water is $5.4 \times 10^{-3} \text{ M s}^{-1}$
- B) The plot of $\ln \frac{C_o}{C_o - x}$ against 't' is a straight line then the reaction may be decomposition of H_2O_2 into H_2O, O_2
- C) The value of rate constant, predicted by the Arrhenius's equation if $T \rightarrow \infty$ is 'A'
- D) The rate constant K_1 and K_2 of two reactions are in the ratio 2:1. The corresponding energies of activation of the reactions will be related as $E_2 > E_1$. If A value is same for two reactions.

39. A certain reaction $A \rightarrow B$ follows the given concentration (Molarity) – time graph. Which of the following statements is/are true?



- A) The reaction is second order with respect to A.
 B) The rate for this reaction at 20 s is approximately $7 \times 10^{-3} \text{ Ms}^{-1}$
 C) The rate for this reaction at 80 s will be $1.75 \times 10^{-3} \text{ Ms}^{-1}$
 D) The [B] is approximately 0.35 M at $t = 60\text{s}$
40. The observed rate of a chemical reaction is substantially lower than the collision frequency. The correct statement(s) is/are
- A) The reactants may not have the required energy
 B) The reactants may not collide in the proper orientation
 C) Collision complex exists for very short time.
 D) Collision frequency over estimates the number of effective collisions
41. The reaction, $mA \rightarrow nB$ occurs according to the energy profile diagram



Which of the following correctly represent the nature of the reaction?

- A) It is an elementary reaction
 B) Its rate determining step is 2nd step
 C) It is an endothermic reaction
 D) Among the intermediates second intermediate is less reactive.

42. For the reaction $A \longrightarrow B$, the rate law expression is $-\frac{d[A]}{dt} = k[A]^{1/2}$. If initial concentration of A is A_0 . Which of the following statement is/are correct regarding this reaction?
- A) The plot of $\sqrt{A}$ against 't' will be non-linear.
- B) The half-life period is $t_{1/2} = \frac{\sqrt{2}(\sqrt{2}-1)}{K} \sqrt{A_0}$
- C) There is a linear decrease of rate of reaction with time
- D) The integrated form of rate expression is $\sqrt{A} = -\frac{Kt}{2} + \sqrt{A_0}$
43. Rate of a chemical reaction $2A(g) \longrightarrow B(g)$ is defined as ' r_B ' = $\frac{1}{V} \frac{dn_B}{dt}$ where, n_B = number of moles of B formed and if C_B = concentration of B then which of the following relation is/are correct?
- A) $r_B = \frac{dC_B}{dt} + \frac{C_B}{v} \frac{dV}{dt}$ (if volume V not constant)
- B) $r_B = \frac{dC_B}{dt}$ (if volume V is constant)
- C) $r_B = \frac{dC_B}{dt} - \frac{C_B}{v} \frac{dV}{dt}$ (if volume V is not constant)
- D) $r_B = \frac{dC_B}{dt}$ (if volume V is not constant)

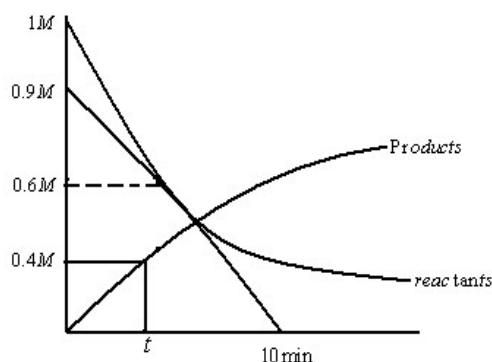
SECTION-II

(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

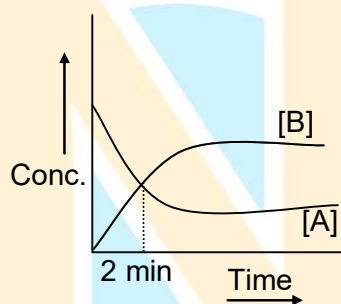
44. $A \rightarrow 2B$ (2^{nd} order reaction)



If the rate of reaction at time 't' is $(0.02)x$. What is "x"?

45. Half-life (T_1) of the first order reaction and half-life (T_2) of the second order reaction are equal. Hence ratio of the rate at the start of the reaction is $\ln X$, the value of X is (If initial concentrations are equal)
46. For the reaction $2NO + H_2 \rightarrow N_2O + H_2O$, value of $\left(-\frac{dp}{dt}\right)$ changes from 1.5 torr s^{-1} to 0.25 torr s^{-1} when pressure is changed from 360 torr to 150 torr with respect to NO keeping pressure of hydrogen to be constant. If pressure of NO is kept constant, it is observed that value of $\left(-\frac{dp}{dt}\right)$ changes from 1.6 torr s^{-1} to 0.8 torr s^{-1} with respect to pressure of hydrogen from 300 torr to 150 torr. Order of the reaction will be [$\log 2 = 0.301$; $\log 3 = 0.477$; $\log 5 = 0.699$; $\log 7 = 0.845$]

47. For a first order reaction $A_{(g)} \rightleftharpoons 3B_{(g)}$ the concentration verses time graph is given below



What is the half-life in minutes (answer to the nearest integer)?

48. The reaction of thioacetamide with water is shown by the equation below.



The rate of reaction is given by the rate law $\text{Rate} = k[H_3O^+][CH_3C(S)NH_2]$. Consider 1 L of solution that is 0.20 M in $CH_3C(S)NH_2$ and 0.15 M in HCl at $25^\circ C$. For each of the changes listed below, the number of changes that decreases the rate of reaction is

- a 4.0 g sample of $NaOH$ is added to the solution.
- 500 ml of water is added to the solution.
- the 0.15 M HCl solution is replaced by 0.15 M acetic acid solution.
- the reaction is carried out at $35^\circ C$ instead of $25^\circ C$.

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q,49, Q,50 and Q,51 by appropriately matching the information given in the three columns of the following table.

Columns 1, 2 and 3 contain the Order of the reaction, value of K and half-life respectively.

	Column-I	Column-II	Column-III
I)	First order reaction	i) $k = \frac{1}{2t} \left[\frac{1}{(a-x)^2} - \frac{1}{a^2} \right]$	P) Half life $\propto \frac{1}{a^2}$
II)	Second order reaction	ii) $k = \frac{1}{t} \left[\frac{1}{a-x} - \frac{1}{a} \right]$	Q) Half life $\propto \frac{1}{a}$
III)	Third order reaction	iii) $k = \frac{x}{t}$	R) Half-life is doubled on doubling the initial concentration.
IV)	Zero order reaction	iv) $k = \frac{1}{t} \log_e \left(\frac{a}{a-x} \right)$	S) 50% reaction takes same time even if concentration is halved or doubled.

49. The correct combination for reaction $A \xrightarrow{2 \times 10^{-3} \text{ min}^{-1}} B$ is:

- A) II-ii-Q B) IV-iv-S C) I-iv-S D) I-iii-R

50. The incorrect combination among the following is:

- A) I-iii-S B) II-ii-Q C) III-i-P D) IV-iii-R

51. Which of the following is the only **wrong** combination?

- A) I-iv-Q B) I-iv-S C) II-ii-Q D) II-ii-Q

Answer Q,52, Q,53 and Q,54 by appropriately matching the information given in the three columns of the following table.

Columns 1, 2 and 3 contain the Order of the reaction, value of K and half-life respectively.

Column-I Compound		Column-II Relationship	Column-III Statement
A)	CH_3COOH ($K_a = 2 \times 10^{-5}$)	(P) $\text{pH} = \frac{1}{2}(\text{p}K_{a1} + \text{p}K_{a2})$	I) Addition of $\text{HCl}(\text{aq})$ can make buffer solution
B)	0.1M NaHCO_3	(Q) $\text{pH} = \frac{1}{2}(\text{p}K_b - \log C)$	II) Increases solubility of $\text{AgCl}(s)$
C)	$\text{NH}_3(\text{aq.})$	(R) $\text{pH} = 7 + \frac{1}{2}(\text{p}K_b + \log C)$	III) pH is independent of concentration.
D)	$\text{CH}_3\text{COONa}(\text{aq.})$	(S) $\text{pOH} = \frac{1}{2}(\text{p}K_b - \log C)$	IV) Addition of $\text{NaOH}(\text{aq})$ can make buffer solution

52. Which of the following option is incorrectly matched?

- A) $A \rightarrow Q \rightarrow \text{IV}$ B) $B \rightarrow S \rightarrow \text{IV}$ C) $C \rightarrow S \rightarrow \text{II}$ D) $C \rightarrow S \rightarrow \text{I}$

53. For NaHCO_3 , the only correct combination is:

- A) $B \rightarrow P \rightarrow \text{III}$ B) $B \rightarrow R \rightarrow \text{IV}$ C) $B \rightarrow Q \rightarrow \text{III}$ D) $B \rightarrow S \rightarrow \text{IV}$

54. For $\text{CH}_3\text{COONa}(\text{aq.})$, the only correct combination is:

- A) $D \rightarrow R \rightarrow \text{III}$ B) $D \rightarrow R \rightarrow \text{I}$ C) $D \rightarrow R \rightarrow \text{IV}$ D) $D \rightarrow S \rightarrow \text{IV}$